



Ismaele Vincent Masiello
Atominstitut, TU Wien
Stadionallee 2, 1020 Vienna, Austria
Email: ismaele.masiello@tuwien.ac.at
Phone: +43-1-58801-141474
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Dear Editors,

Please find enclosed our manuscript entitled “Direct extraction of anomalous weak values in a generalized Mach–Zehnder interferometer from intensity measurements”, which we would like to submit for publication in *Nature Communications*. Quantumness is a central resource for fundamental research and modern quantum technologies, such as quantum computing and communication. Therefore, identifying the boundaries between quantumness and classicality is of high relevance. Weak values represent a valuable tool to probe such boundaries, but their experimental determination is challenging.

Here, we introduce and experimentally demonstrate a more direct method to extract weak values. Our scheme is verified using a neutron Mach–Zehnder interferometer, a matter-wave platform allowing for the investigation of one of the fundamental elements of quantum mechanics—the two-level system—under conditions that challenge classical interpretations: a single massive particle in a superposition of two macroscopically separated paths. We show that our approach achieves high accuracy while significantly reducing experimental resources and measurement time. Furthermore, we directly observe anomalous weak values associated with “negative probabilities”, known as quasiprobabilities, reflecting the nonclassical behavior of the quantum system.

The generality and simplicity of our method make it applicable to a wide range of platforms, including photonic and atomic systems, providing a flexible and experimentally accessible tool for probing nonclassicality. These results open new avenues for experimentally investigating the quantum–classical boundary and provide practical advantages for future studies. We believe that the presented work will be of broad interest to the readership of *Nature Communications*, particularly in quantum foundations, quantum measurement theory, and emerging quantum technologies.

We confirm that this work is original and unpublished, and it is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission to *Nature Communications*.

We suggest the following reviewers

Micheal J.W. Hall, <i>Australian National University</i>	Justin Dressel, <i>Chapman University</i>
Sandu Popescu, <i>University of Bristol</i>	Dmitry Pushin, <i>University of Waterloo</i>
W. Michael Snow, <i>Indiana University Bloomington</i>	

Best Regards,

Ismaele Vincent Masiello*, Andreas Dvorač, Hartmut Lemmel, Stephan Sponar, and Yuji Hasegawa

(*corresponding author)